

Taylor Tube Method for Validated IVP[★]

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Abstract. We recently introduced a novel architecture for the design of validated IVP algorithms. This architecture forms the basis of our complete validated algorithm for IVP. A key subroutine in our algorithm is the Euler Tube: it gave a technique for refining end- and full-enclosures and is also key to deriving a complexity bound of our IVP solver. In this paper, we generalize it to Taylor Tube of degree $p \geq 1$. We stress that our Taylor Tube technique is distinct from the well-known Taylor Models. As expected, higher-degree Taylor tubes improve accuracy. Our experiments further show that it can also lead to an overall speedup when combined with bisection.

Keywords: initial value problem · validated numerics · complete algorithm · Taylor tube method · reachability problem · logarithmic norm

1 Introduction

Consider the initial value problem (IVP) for an autonomous ODE of the form

$$\mathbf{x}' = \mathbf{f}(\mathbf{x}). \quad (1)$$

where $\mathbf{f} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is C^k -continuous (for some global constant $k \geq 1$). For $B_0 \subseteq \mathbb{R}^n$ ($n \geq 1$) and $h > 0$, let $\text{IVP}_{\mathbf{f}}(B_0, h)$ denote the set of solutions $\mathbf{x} : [0, h] \rightarrow \mathbb{R}^n$ that satisfy (1) and $\mathbf{x}(0) \in B_0$. When the initial set is a singleton, $B_0 = \{\mathbf{q}_0\}$, we abbreviate $\text{IVP}_{\mathbf{f}}(\{\mathbf{q}_0\}, h)$ as $\text{IVP}_{\mathbf{f}}(\mathbf{q}_0, h)$. We say (B_0, h) is valid (or $\mathbf{f}$ -valid) if for each $\mathbf{p}_0 \in B_0$, some $\mathbf{x} \in \text{IVP}_{\mathbf{f}}(B_0, h)$ with $\mathbf{x}(0) = \mathbf{p}_0$ has unique-existence.¹

For a valid (B_0, h) , let

$$\begin{aligned} \text{End}_{\mathbf{f}}(B_0, h) &:= \{\mathbf{x}(h) : \mathbf{x} \in \text{IVP}_{\mathbf{f}}(B_0, h)\}, \\ \text{Image}_{\mathbf{f}}(B_0, h) &:= \{\mathbf{x}(t) : \mathbf{x} \in \text{IVP}_{\mathbf{f}}(B_0, h), t \in [0, h]\}. \end{aligned} \quad (2)$$

We fix $\mathbf{f}$ in this paper; so the subscript $\mathbf{f}$ may be omitted above. By an **end-closure** and a **full-enclosure** for (B_0, h) we mean any sets E and F that contain

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¹ i.e., it exists, and it is unique.

30 $\text{End}(B_0, h)$ and $\text{Image}(B_0, h)$, respectively. Two central problems in the appli-
 31 cations of IVP are to compute such end- and full-enclosures. Such enclosure
 32 questions are closely related to reachability analysis studied in control theory
 33 and its applications [5, 13]. The literature on validated algorithms for IVP has a
 34 history of over 50 years going back to Moore, and are of course based on interval
 35 methods. See the surveys of Corliss [4] and Nedialkov et al. [9].

36 We formulate the following reachability problem, called the **End Cover**
 37 **Problem:** Given (f, B_0, h, ε) ($h, \varepsilon > 0$) where B_0 is a box, to compute a fi-
 38 nite set $\mathcal{C} = \{B_1, \dots, B_m\}$ of boxes such that

$$\text{End}_f(B_0, h) \subseteq \left(\bigcup \mathcal{C} \right) \subseteq \text{End}_f(B_0, h) \oplus [-\varepsilon, \varepsilon]^n \quad (3)$$

39 where $\bigcup \mathcal{C} = \bigcup_{i=1}^m B_i$ and $A \oplus B$ denotes the Minkowski sum of Euclidean sets
 40 $A, B \subseteq \mathbb{R}^n$. Any $\mathcal{C}$ that satisfies (3) is called an ε -**cover** of $\text{End}_f(B_0, h)$.

41 Recently, we introduced an algorithm [15, 16] for this End Cover Problem.
 42 It represented the first complete validated IVP algorithm. By “complete” we
 43 mean that our algorithm is a halting algorithm. Most validated algorithms are
 44 only partially correct² in the sense that, the output is correct *if it halts*. Halt-
 45 ing is especially challenging in IVP problems, partly because the error typically
 46 grows exponentially with time. We note some unique features in our solution: (1)
 47 We guarantee that the “over-approximation” of $\mathcal{C}$ is within any user-specified ε
 48 bound. Rigorous accuracy control generally requires a covering approach rather
 49 than a single enclosing box. Even the exact interval hull of the reachable set may
 50 be wider than the prescribed tolerance, simply because of the geometry of the
 51 set. Thus the End-Cover Problem seeks a finite cover by boxes, instead of one
 52 global box enclosure. (2) Our algorithm is fully automatic; it does not require
 53 hyper-parameters, such as a minimum step size or tolerance bounds, which are
 54 commonly used in such algorithms. (3) In preliminary implementations, our al-
 55 gorithm has proved to be practical. It is reasonable to expect that our algorithm
 56 takes a performance hit in order to ensure completeness. Although complete cer-
 57 tification usually entails computational overhead, our algorithm often performs
 58 better than current validated IVP software [14, 3].

59 Our algorithm [15, 16] introduced a novel architecture for IVP algorithms
 60 based on the **scaffold** framework. Within this framework, many of our original
 61 subroutines could be swapped. One of these subroutines is called **Euler tube**.
 62 In this paper, we generalize it into a **Taylor tube** method, based on using the
 63 first $p \geq 1$ terms of a Taylor expansion. Call p the **tube degree**. The case of
 64 Euler tube corresponds to $p = 1$. In the framework of validated IVP algorithms,
 65 our Taylor Tube can be viewed as a technique for implementing stepB in the
 66 stepA/stepB iteration (see section 3 below or [15]). Our experiments show that
 67 a higher degree for p leads to tighter enclosures (not a surprise) but surprisingly,
 68 they can also lead to overall speedup. We remark that such tube techniques
 69 are distinct from the well-known Taylor model [1], and they are critical for our
 70 ability to prove complexity bounds for our halting algorithm.

² This is a standard terminology in theoretical computer science.

2 Taylor Tube

Notations. Most of the following notations are taken from [15,16]. Vector variables such as $\mathbf{x} = (x_1, \dots, x_n)$ or $\mathbf{p}$ are in bold font. For $\mathbf{p} \in \mathbb{R}^n$ and $r \geq 0$, let $Ball_{\mathbf{p}}(r) := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{p}\|_2 \leq r\}$ denote the Euclidean ball centered at $\mathbf{p}$ with radius r . For $r \geq 0$, define the axis-aligned box $Box_{\mathbf{p}}(r) := \mathbf{p} + [-r, r]^n$. When $\mathbf{p} = \mathbf{0}$, we simply write $Box(r) := Box_{\mathbf{0}}(r) = [-r, r]^n$. For a box $B = \prod_i [\underline{b}_i, \bar{b}_i]$, let $m(B) := \left(\frac{\underline{b}_1 + \bar{b}_1}{2}, \dots, \frac{\underline{b}_n + \bar{b}_n}{2}\right)_i$, $w_{\max}(B) := \max_i (\bar{b}_i - \underline{b}_i)$. For a square matrix $A \in \mathbb{R}^{n \times n}$ and the induced matrix q -norm, the **logarithmic q -norm** is

$$\mu_q(A) := \lim_{h \rightarrow 0+} \frac{\|I + hA\|_q - 1}{h}.$$

We focus on $q = 2$ and write $\mu(A) = \mu_2(A)$ as the logNorm. For any $B \subseteq \mathbb{R}^n$, define

$$\mu(J_{\mathbf{f}}(B)) := \sup \{\mu(J_{\mathbf{f}}(\mathbf{p})) : \mathbf{p} \in B\}.$$

Call $\bar{\mu}$ a **logNorm-bound** on B if $\bar{\mu} \geq \mu(J_{\mathbf{f}}(B))$. Unlike classical norms, logarithmic norms can be negative. We have this basic result:

Lemma 1. *Let $\mathbf{x}_1, \mathbf{x}_2 \in IVP(Ball_{\mathbf{p}_0}(r), h)$ and assume that $\bar{\mu}$ is a logNorm bound on F where F is any full-enclosure for $(Ball_{\mathbf{p}_0}(r), h)$. For all $t \in [0, h]$,*

$$\|\mathbf{x}_1(t) - \mathbf{x}_2(t)\|_2 \leq \|\mathbf{x}_1(0) - \mathbf{x}_2(0)\| e^{\bar{\mu}t}. \quad (4)$$

This lemma is just a special case of the logNorm version of the **Fundamental IVP Inequality** ([11,6]).

Tubular region of a trajectory. Fix a trajectory $\mathbf{x} = IVP(\mathbf{q}_0, h)$ where $\mathbf{x}(0) = \mathbf{q}_0$. For any $\delta > 0$, the δ -**tube** of $\mathbf{x}$ is the set

$$\text{Tube}_{\delta}(\mathbf{x}) := \{(t, \mathbf{p}) : 0 \leq t \leq h, \|\mathbf{p} - \mathbf{x}(t)\|_2 \leq \delta\} \quad (\subseteq [0, h] \times \mathbb{R}^n).$$

An arbitrary curve $C : [0, \infty) \rightarrow \mathbb{R}^n$ lies in the δ -tube of $\mathbf{x}$ if $(t, C(t)) \in \text{Tube}_{\delta}(\mathbf{x})$ for all $t \in [0, h]$. For $p \geq 1$, the Taylor polynomial of $\mathbf{f}$ of degree p at any $\mathbf{q} \in \mathbb{R}^n$ is defined to be

$$T_{\mathbf{q}}^p(t) := \sum_{i=0}^p t^i \mathbf{f}^{[i]}(\mathbf{q}) \quad (5)$$

where ³

$$\mathbf{f}^{[i]}(\mathbf{x}) := \begin{cases} \mathbf{x} & \text{if } i = 0, \\ \frac{1}{i} (J_{\mathbf{f}^{[i-1]}} \bullet \mathbf{f})(\mathbf{x}) & \text{else} \end{cases}$$

where $J_{\mathbf{g}}$ denotes the Jacobian of $\mathbf{g}$ and $\bullet$ is matrix-vector product. The **Taylor sequence** of $\mathbf{q}_0$ with step size $\bar{h} > 0$ is the infinite sequence of points $\mathbf{q}_0, \mathbf{q}_1, \mathbf{q}_2 \dots$

³ We may call this the normalized i -th Taylor differential coefficient of $\mathbf{f}$, where “differential” here emphasizes that it is not the usual coefficient of polynomial.

94 where $\mathbf{q}_{i+1} := T_{\mathbf{q}_i}^p(\bar{h})$ for all $i \geq 0$. This sequence defines the piece-wise continu-
 95 ous Taylor curve $T_{\mathbf{q}_0, \bar{h}}^p(t)$ of step size $\bar{h}$ where

$$T_{\mathbf{q}_0, \bar{h}}^p(t) := T_{\mathbf{q}_i}^p(t - i\bar{h}), \quad (t \geq 0, i = \lfloor t/\bar{h} \rfloor). \quad (6)$$

96 We could similarly define the δ -tube of the Taylor curve $T_{\mathbf{q}_0, \bar{h}}^p(t)$. A natural
 97 question is whether this tube contains the true trajectory $\mathbf{x}$. The next lemma
 98 answers this question in the affirmative, under an explicit upper bound on the
 99 Taylor curve with step-size $\bar{h}$. The lemma is a consequence of the fundamental
 100 IVP inequality (Lemma 1); its proof can be found in [17].

101 **Lemma 2 (Taylor Tube Step Size Bound).** *Let $\mathbf{x} = \text{IVP}(\mathbf{q}_0, h)$ and B_1
 102 be a full enclosure of $(\mathbf{q}_0, h)$. Assume that $\bar{\mu}$ is a logNorm-bound on B_1 and
 103 $\bar{M} \geq \|\mathbf{f}^{[p+1]}(B_1)\|$. For any $\delta > 0$, if $\bar{h}$ satisfies*

$$\bar{h} \leq h_p^{\text{Taylor}}(h, \bar{M}, \bar{\mu}, \delta) := \begin{cases} \left(\frac{\bar{\mu} \delta}{\bar{M}(e^{\bar{\mu} h} - 1)} \right)^{1/p} & \text{if } \bar{\mu} > 0, \\ \left(\frac{\delta}{\bar{M} h} \right)^{1/p} & \text{if } \bar{\mu} = 0, \\ \min \left\{ \left(\frac{\bar{\mu} \delta}{2\bar{M}(e^{\bar{\mu} h} - 1)} \right)^{1/p}, -\frac{1}{\bar{\mu}} \right\}, & \text{if } \bar{\mu} < 0, \end{cases} \quad (7)$$

104 and if $\text{Image}_{\mathbf{f}}(\mathbf{q}_i, \bar{h}) \subseteq B_1$ (for all $i = 0, \dots, \lfloor h/\bar{h} \rfloor$) then the δ -tube of $\mathbf{x}$
 105 contains the Taylor curve $T_{\mathbf{q}_0, \bar{h}}^p(t)$ of step size $\bar{h}$.

106 3 Overview of the End-Cover Algorithm

107 Most validated IVP algorithms are based on two key subroutines which we call⁴
 108 **StepA** and **StepB**. Given a box E_0 , we may call **StepA** and **StepB** in sequence to
 109 transform E_0 into an admissible triple and then admissible quad:

$$E_0 \xrightarrow{\text{StepA}} (E_0, h, F_1) \xrightarrow{\text{StepB}} (E_0, h, F_1, E_1). \quad (8)$$

110 In general, if (B_0, h) is valid and E, F are end- and full-enclosures for (B_0, h) then
 111 (B_0, h, F) (resp., (B_0, h, F, E)) is called an **admissible triple** (resp., **quad**).
 112 For $i \geq 1$, we transform $E_{i-1} \rightarrow (E_{i-1}, h_i, F_i, E_i)$ which is an admissible quad.
 113 Note that E_m ($m \geq 1$) is an end-enclosure for (E_0, t_m) where $t_m = \sum_{i=1}^m h_i$.
 114 We need admissible triples because global unique-existence of solutions must be
 115 built-up incrementally in small time steps which are justified by the classical
 116 Picard-Lindelöf theorem.

Next, we introduce the **scaffold data structure** $\mathcal{S}$ to store such quads. If
 there are m quads, we use these dot-notations for the various parameters such
 as

$$\mathcal{S}.m, \quad \mathcal{S}.E_0, \quad \mathcal{S}.t, \quad \mathcal{S}.E_m$$

⁴ **StepA** in [15], has two extra arguments $h, \varepsilon \geq 0$, which is defaulted to $h = \infty, \varepsilon = 0$ here.

for the number (m) of quads, the initial box (E_0) of the first quad, the total time (t_m), and the end-enclosure (E_m) at time t_m . Each quad represents a **stage** of $\mathcal{S}$. The goal is to ensure that (a) $\mathcal{S}.t = H$ and (b) $w_{\max}(\mathcal{S}.E_m) < \varepsilon$ where H, ε are the input arguments. If (a) fails, we call a subroutine $\mathcal{S}.\text{Extend}(H)$ that adds a new stage (using **StepA/StepB**). If (b) fails, we call $\mathcal{S}.\text{Refine}(\varepsilon)$ to reduce $w_{\max}(\mathcal{S}.E_m)$. However, condition (b) may not be possible without subdividing $\mathcal{S}.E_0$ into subboxes. After subdividing E_0 , each subbox $E'_0 \subseteq E_0$ becomes the initial box of a new m -stage scaffold that inherits all the quads of $\mathcal{S}$. The basic operation of $\mathcal{S}.\text{Refine}(\varepsilon)$ proceeds in **phases**: each phase refines all m stages (from stage $i = 1, \dots, m$ in this order). Refining stage i means to convert the i th quad

$$(E_{i-1}, h_i, F_i, E_i) \xrightarrow{\text{refine}} (E_{i-1}, h_i, F'_i, E'_i) \quad (9)$$

to a new admissible quad with $F'_i \subseteq F_i$ and $E'_i \subseteq E_i$. If $Q_i = (E_{i-1}, h_i, F_i, E_i)$ then we refine Q_i by generating 2^{ℓ_i} **mini-steps** (with uniform step-size $h_i 2^{-\ell_i}$) between Q_{i-1} and Q_i :

$$Q_{i-1} \longrightarrow Q_i^{(1)} \longrightarrow Q_i^{(2)} \longrightarrow \dots \longrightarrow Q_i^{(2^{\ell_i}-1)} \longrightarrow Q_i. \quad (10)$$

We provide two subroutines **Bisect** and the **TaylorTube** for doing this refinement: **Bisect** ([15, section 6.3]) simply calls **StepB** twice with half the current mini-step size. **TaylorTube** is similar to **EulerTube** ([15, section 6.3]) except that we use the Taylor sequence $\mathbf{q}_0, \mathbf{q}_1, \dots$ and the step size bound h_p^{Taylor} in (7). When the mini-step size is smaller than the bound in Lemma 2, we can speed up by using the Taylor tube method described in the next section.

See Appendix XXX for the pseudo-code of End-Enclosure Algorithm.

4 Refining End- and Full-Enclosures with Taylor Tubes

We now show how the Taylor Tube method can be combined with **StepB**. One of the best previous methods for **StepB** is the Direct Method in [10,9]:

$$\begin{aligned} \text{Direct}(E_0, h, F_1) &= \underbrace{\sum_{i=0}^{k-1} h^i \mathbf{f}^{[i]}(\mathbf{m}(E_0))}_{\text{Point}} + \underbrace{h^k \mathbf{f}^{[k]}(F_1)}_{\text{Point Error}} + \underbrace{\left(\sum_{i=0}^{k-1} h^i J_{\mathbf{f}^{[i]}}(E_0) \right) \bullet (E_0 - \mathbf{m}(E_0))}_{\text{Range Enclosure}} \\ &= T_{\mathbf{m}(E_0)}^{k-1}(h) + \text{PE}(h, F_1) + \text{RE}(h, E_0) \end{aligned} \quad (11)$$

Note that $\text{Direct}(E_0, h, F_1)$ and $\text{Direct}(E_0, [0, h], F_1)$ produce (respectively) an end-enclosure (E_1) and a refinement of the full-enclosure (F_1) for (E_0, h) . Our next theorem provides a new method for refining F_1 . We state the result here and refer the reader to [17] for the proof.

Theorem 1 (Taylor Tube Enclosures).

Consider an admissible triple (E_0, h, F_1) where $E_0 = \text{Ball}_{\mathbf{q}_0}(r_0)$. Let $\delta > 0$, $\bar{h} \leq \min\{h, h_p^{\text{Taylor}}(h, \bar{M}, \bar{\mu}, \delta)\}$, $\bar{\mu}$ is a $\log\text{Norm}$ -bound on F_1 and $\bar{M} \geq \|\mathbf{f}^{[p+1]}(F_1)\|$. If

$$\text{Image}(\mathbf{q}_i, \bar{h}) \subseteq F_1 \quad (\text{for all } i = 0, \dots, \lfloor h/\bar{h} \rfloor) \quad (12)$$

150 then the following holds:

- 151 (a) An end-enclosure for $IVP(E_0, h)$ is given by $Ball_{T_{\mathbf{q}_0, \bar{h}}^p(h)}(r_0 e^{\bar{\mu}h} + \delta)$.
 152 (b) A full-enclosure for $IVP(E_0, h)$ is given by $T_{\mathbf{q}_0, \bar{h}}^p([0, h]) \pm [-r, r]^n$, where $r =$
 153 $\max_{h' \in [0, h]} (r_0 e^{\bar{\mu}h'} + \delta) = \delta + \max\{r_0, r_0 e^{\bar{\mu}h}\}.$

154 We can combine **Direct** with Theorem 1 to define a tighter end-enclosure $E_1(h)$
 155 defined as follows:

$$E_1(h) := \begin{cases} \text{Direct}(E_0, h, F_1) \cap Ball_{T_{\mathbf{q}_0}^p(h)}(r_0 e^{\bar{\mu}h} + \delta), & \text{if } p \neq k-1, \\ T_{\mathbf{m}(E_0)}^{k-1}(h) + (\text{PE}(h, F_1) \cap \text{Box}(\delta)) + (\text{RE}(h, E_0) \cap \text{Box}(r_0 e^{\bar{\mu}h})), & \text{if } p = k-1. \end{cases} \quad (13)$$

156 **Remark:** When $p = k-1$, we have

$$T_{\mathbf{q}_0}^p(h) = T_{\mathbf{m}(\mathbf{E}_0)}^{k-1}(h),$$

157 which is a single point. In this case, we use the formula in (13) with $p = k-1$ to
 158 obtain a tighter end-enclosure. This refinement follows from (11) together with
 159 the elementary inclusion

$$a + (S_1 \cap S'_1) + (S_2 \cap S'_2) \subseteq (a + S_1 + S_2) \cap (a + S'_1 + S'_2),$$

160 where $a \in \mathbb{R}^n$ is a point, $S_1, S_2, S'_1, S'_2 \subseteq \mathbb{R}^n$ are sets, and all additions are
 161 understood as Minkowski sums.

162 Furthermore, $E_1([0, h]) \cap F_1$ gives a tighter full-enclosure than F_1 . We com-
 163 pute $E_1([0, h])$ using the formula (13) with the argument $[0, h]$ instead of h .

164 Our Taylor Tube algorithm can check (12) by using this inclusion

$$T_{\mathbf{q}_i, \bar{h}}^p([0, \bar{h}]) + [0, \bar{h}]^{p+1} \mathbf{f}^{[p+1]}(F_1) \subseteq F_1. \quad (14)$$

165 Note that $T_{\mathbf{q}_i, \bar{h}}^p([0, \bar{h}])$ and $\mathbf{f}^{[p+1]}(F_1)$ have already been computed when com-
 166 puting $E_1([0, \bar{h}])$ and $\bar{M}$.

167 5 Experiments

168 We present experimental evidence for the effectiveness of our Taylor tube tech-
 169 niques, using the list of ODE examples in Table 1.

170 5.1 One Step Comparison of Taylor Tube formulas (Table 2)

171 In Table 2, for each example and for any given admissible (E_0, H, F_1) (which is
 172 obtained from **StepA**), we first compute $\bar{h} = \min(H, h^{\text{Taylor}}(H, \bar{M}, \mu(J_{\mathbf{f}}(F_1)), \delta))$
 173 (see (7)). In this table, we investigate the dependence of $\bar{h}$ on δ : $\bar{h} = \bar{h}(\delta)$.
 174 Then we compute $\mu := \mu_2(J_{\mathbf{f}}(\text{Direct}(E_0, [0, \bar{h}], F_1)))$ and the full-enclosure of

Table 1. List of IVP Problems

Eg*	Name	$f(x)$	Parameters	Box B_0	Reference
Eg1	Volterra	$\begin{pmatrix} ax(1-y) \\ -by(1-x) \end{pmatrix}$	$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$	$Box_{(1,3)}(0.1)$	[8], [1, p.13]
Eg2	Van der Pol	$\begin{pmatrix} y \\ -c(1-x^2)y-x \end{pmatrix}$	$c = 1$	$Box_{(-3,3)}(0.1)$	[1, p.2]
Eg3	Asymptote	$\begin{pmatrix} x^2 \\ -y^2 + 7x \end{pmatrix}$	N/A	$Box_{(-1.5,8.5)}(0.01)$	N/A
Eg4	Lorenz	$\begin{pmatrix} \sigma(y-x) \\ x(\rho-z)-y \\ xy-\beta z \end{pmatrix}$	$\begin{pmatrix} \sigma \\ \rho \\ \beta \end{pmatrix} = \begin{pmatrix} 10 \\ 28 \\ 8/3 \end{pmatrix}$	$Box_{(15,15,36)}(0.001)$	[1, p.11]
Eg5	Rössler	$\begin{pmatrix} -y-z, x+ay, b+z(x-c) \end{pmatrix}$	$(a, b, c) = (0.2, 0.2, 5.7)$	$Box_{(1,2,3)}(0.1)$	[12]

175 IVP($E_0, \bar{h}$) using, respectively, the formulas (11) and (13). We compare the rel-
 176 ative effectiveness of the 2 formulas using the ratio

$$\sigma := \frac{vol(\text{Direct}(E_0, [0, \bar{h}], F_1))}{vol(E_1([0, \bar{h}]))}, \quad (15)$$

177 where $vol(\cdot)$ is the volume of a box. Thus $\sigma > 1$ implies an improvement due
 178 to Taylor tube. In the table, we choose different values of δ and Taylor order k
 179 (with $p = k - 1$).

180 From Table 2, we see that σ (last column) is generally > 1 implying the
 181 effectiveness of Taylor tube. Moreover, increasing the tube degree causes the step
 182 size h_p^{Taylor} to increase (see the $\bar{h}$ column). When $\bar{h}$ is capped at the value $\bar{h} = H$,
 183 increasing the tube degree can still slightly improve σ , although in this one-step
 184 experiment the improvement is often below the two-decimal resolution shown in
 185 the table. The significance of this effect is that even small local improvements
 186 may accumulate exponentially over many integration steps.

187 5.2 Global Experiment (Table 3)

188 In contrast to the previous one-step experiments, we now do global experiments
 189 using the full algorithm. Table 3 shows the running times of our End-Cover
 190 Algorithm on each example using varying tube degrees (1, 3, 5, 7, 19). The order
 191 k is fixed at 20. We also record the size $|\mathcal{C}|$ of the end-cover. But in each case,
 192 we show two numbers, representing the standard End-Cover Algorithm using
 193 **Bisect (B)** and its variant (**NoB**) in which the **Bisect** subroutine is turned
 194 off. The effect of the tube degree is coupled with other certification mechanisms.
 195 When certification fails, the algorithm may reduce the step size or bisect the
 196 current initial subbox. Moreover, the Taylor-based enclosure is intersected with
 197 the enclosure produced by **TaylorTube**, which is based on a logarithmic-norm
 198 version of the fundamental IVP inequality. This bound is often conservative, but
 199 not uniformly worse than the Taylor bound, especially for small step sizes. Hence
 200 the observed performance depends on the relative sharpness of these bounds on
 201 each subproblem. The **NoB** variant removes initial-box subdivision and helps
 202 make the influence of the tube degree more visible.

Table 2. Comparison of Full-Enclosures from $Direct(E_0, [0, \bar{h}], F_1)$ and $E_1([0, \bar{h}])$.

Eg	E_0	H	F_1	δ	Taylor order k	$\bar{h}$	μ	σ
Volterra	$(1.0, 3.0) \pm (0.1, 0.1)$	0.1	$(0.75, 3.0) \pm (0.36, 0.20)$	0.1	1	0.0144	0.1378	1.00
					3	0.1	0.4386	1.03
					7	0.1	0.4171	1.07
					20	0.1	0.4169	1.07
				0.01	1	0.00201	0.1048	1.09
					3	0.0525	0.2398	1.30
					7	0.1	0.36	1.69
					20	0.1	0.36	1.69
				0.001	1	0.00020	0.1005	1.00
					3	0.0275	0.17	1.23
					7	0.1	1.9	1.85
					20	0.1	0.35	1.85
Van der Pol	$(-3.0, 3.0) \pm (0.1, 0.1)$	0.05	$(-2.9, 2.4) \pm (0.19, 0.71)$	0.1	1	0.0035	0.6314	1.10
					3	0.0481	1.1	1.69
					7	0.05	1.2	1.76
					20	0.05	1.2	1.76
				0.01	1	0.00041	0.60	1.22
					3	0.0239	0.83	1.34
					7	0.05	1.1	2.02
					20	0.05	1.2	2.02
				0.001	1	4.1×10^{-5}	0.60	1.05
					3	0.0114	0.70	1.16
					7	0.05	1.0	2.09
					20	0.05	1.1	2.09
Asymptote	$(-1.50, 8.50) \pm (0.01, 0.01)$	0.04	$(-1.5, 6.8) \pm (0.059, 1.7)$	0.1	1	0.00117	0.1167	1.02
					3	0.0246	2.3	1.42
					7	0.04	3.5	1.97
					20	0.04	3.5	1.97
				0.01	1	0.00012	0.030	1.01
					3	0.0116	1.0	1.22
					7	0.04	3.4	2.14
					20	0.04	3.4	2.15
				0.001	1	1.2×10^{-5}	0.021	1.00
					3	0.0054	0.47	1.10
					7	0.0305	2.8	1.82
					20	0.04	3.3	2.24
Lorenz	$(15.000, 15.000, 36.000) \pm (0.001, 0.001, 0.001)$	0.027	$(15, 13, 38) \pm (0.26, 2.2, 1.8)$	0.1	1	0.000508	2.1	1.000
					3	0.0151	2.9	1.000
					7	0.027	3.3	1.000
					20	0.027	3.3	1.000
				0.01	1	5.3×10^{-5}	2.1	1.000
					3	0.00719	2.5	1.000
					7	0.027	3.3	1.000
					20	0.027	3.3	1.000
				0.001	1	5.4×10^{-6}	2.1	1.000
					3	0.00387	2.3	1.000
					7	0.0248	3.2	1.001
					20	0.027	3.3	1.001
Rössler	$(1.0, 2.0, 3.0) \pm (0.1, 0.1, 0.1)$	0.1	$(0.74, 2.1, 2.3) \pm (0.37, 0.18, 0.85)$	0.1	1	0.00588	0.3155	1.00
					3	0.1	0.71	1.62
					7	0.1	0.70	1.73
					20	0.1	0.70	1.73
				0.01	1	0.00063	0.30	1.00
					3	0.0492	0.48	1.32
					7	0.1	0.65	2.02
					20	0.1	0.65	2.02
				0.001	1	6.3×10^{-5}	0.29	1.00
					3	0.0257	0.39	1.17
					7	0.1	0.64	2.12
					20	0.1	0.64	2.12

For each example, we add a special row whose tube is degree “19 (**NoT**)”. This is our End-Cover Algorithm with **TaylorTube** turned off. Comparing the row for degree 19 with this special row is extremely insightful: we get a triple of timing in seconds. For example, for Volterra, the triple is (21.7s, **7.06s**, 19.3s), corresponding to (**T+NoB**, **T+B**, **NoT+B**). Inevitably, the middle value is smallest, showing that the combination of **Bisect** and **TaylorTube** is better than either one alone.

Observe that the use of **Bisect** not only improves the overall efficiency, but also tends to reduce the size of the end-covers (this is seen by comparing the numbers in the **NoB** and **B** columns).

Remark: compared to other tube degrees, the timing for degree 19 is artificially lower since it matches the order $k = 20$ used in **StepB** of (13). This allows the Taylor-tube computation to reuse quantities computed in **StepB** such as the point-error term.

Table 3. Performance of End-Cover algorithm under different tube degrees.

Model	T	ϵ	Tube degree p	Cover size $ C $		Time (s)	
				NoB	B	NoB	B
Volterra	2	0.01	1	5968	1024	47.5	8.05
			3	5758	1024	44.5	7.82
			5	5758	1024	46.2	10.6
			7	5719	1024	47.3	11.0
			19	2461	1024	21.7	7.06
			19 (NoT)	-	1024	-	19.3
Van der Pol	2	0.01	1	13219	4174	75.42	35.58
			3	9196	4195	60.88	35.99
			5	6385	4111	89.94	34.74
			7	4789	4102	99.55	34.48
			19	4732	4098	44.70	34.10
			19 (NoT)	-	4096	-	35.28
Asymptote	1	0.01	1	4933	4096	53.05	52.08
			3	4936	4096	56.65	53.64
			5	4930	4096	60.55	53.90
			7	4705	4096	62.13	53.65
			19	4642	4096	62.61	51.14
			19 (NoT)	-	4096	-	51.846
Lorenz	1	1.0	1	246	8	10.6	0.293
			3	246	8	14.0	0.296
			5	246	8	16.2	0.302
			7	246	8	18.6	0.300
			19	211	8	36.4	0.274
			19 (NoT)	-	8	-	1.75
Rössler	2	1.0	1	8	8	0.146	0.140
			3	8	8	0.165	0.166
			5	8	8	0.188	0.185
			7	8	8	0.199	0.210
			19	8	8	0.138	0.111
			19 (NoT)	-	8	-	0.370

6 Conclusion

The introduction of Taylor-tube methods provides a new tool for validated IVP computation. Our experiments show that, within the End-Cover Algorithm, these methods can be effective in producing certified enclosures. A natural direction for future work is to combine our approach with Taylor-model data structures [7, 2].

A Appendix A: Proofs

Proof of Lemma 2.

For simplicity, assume that $h = m\bar{h}$ for some integer m , and write

$$t_i = i\bar{h}, \quad i = 0, \dots, m.$$

The proof below is easily adjusted if the last step size is $< \bar{h}$.

For each $i = 0, \dots, m-1$, let

$$\mathbf{q}_i := T_{\mathbf{q}_0, \bar{h}}^p(t_i), \quad g_i := \|\mathbf{q}_i - \mathbf{x}(t_i)\|$$

and

$$\mathbf{x}_i : [0, \bar{h}] \rightarrow \mathbb{R}^n$$

denotes the exact solution of $\mathbf{x}_i'(s) = \mathbf{f}(\mathbf{x}_i(s))$, $\mathbf{x}_i(0) = \mathbf{q}_i$. By hypothesis, $\text{Image}(\mathbf{q}_i, \bar{h}) \subseteq B_1$. Since B_1 is a full enclosure of $(\mathbf{q}_0, h)$, we also have $\mathbf{x}([t_i, t_{i+1}]) \subseteq B_1$. Hence, we can invoke Lemma 1 to bound the error between $\mathbf{x}$ and $\mathbf{x}_i$ over their common time interval:

$$\|\mathbf{x}_i(s) - \mathbf{x}(t_i + s)\| \leq \|\mathbf{q}_i - \mathbf{x}(t_i)\| e^{\bar{\mu}s} = g_i e^{\bar{\mu}s}. \quad (16)$$

Next, we bound the error $E(t)$ between the Taylor curve and $\mathbf{x}$ at time $t \in [0, h]$ as follows:

$$\begin{aligned} E(t) &:= \|T_{\mathbf{q}_0, \bar{h}}^p(t) - \mathbf{x}(t)\| \\ &= \|T_{\mathbf{q}_i}^p(s) - \mathbf{x}(t_i + s)\| \\ &\quad (\text{where } i = \lfloor t/\bar{h} \rfloor, t = s + t_i \text{ and } 0 \leq s < \bar{h}) \\ &\leq \|T_{\mathbf{q}_i}^p(s) - \mathbf{x}_i(s)\| + \|\mathbf{x}_i(s) - \mathbf{x}(t_i + s)\| \\ &\quad (\text{by triangular inequality and } \mathbf{q}_i = \mathbf{x}_i(0) = \mathbf{x}(t_i)) \\ &\leq \|T_{\mathbf{q}_i}^p(s) - \mathbf{x}_i(s)\| + g_i e^{\bar{\mu}s} \\ &\quad (\text{by (16)}) \\ &= \|f^{[p+1]}(\mathbf{q}_i)\xi^{p+1}\| + g_i e^{\bar{\mu}s} \\ &\quad (\text{Taylor remainder, } 0 \leq \xi \leq s) \\ &\leq \bar{M}s^{p+1} + g_i e^{\bar{\mu}s} \\ &\quad (\mathbf{q}_i \in B_1 \text{ and } \|f^{[p+1]}(B_1)\| \leq \bar{M}). \end{aligned}$$

Let $\phi(s)$ be the last expression in this derivation:

$$\phi_i(s) := \bar{M}s^{p+1} + g_i e^{\bar{\mu}s}. \quad (17)$$

Note that $\phi_i(s)$ is convex, being the sum of convex functions $\overline{M}s^{p+1}$ and $g_i e^{\overline{\mu}s}$, and so its extremal values are obtained at $\phi_i(0)$ and $\phi_i(\overline{h})$. Let $\phi : [0, h] \rightarrow \mathbb{R}$ where $\phi(t) := \phi_i(s)$ if $t = s + t_i$, $i = \lfloor s/\overline{h} \rfloor$. Thus ϕ is an upper bound on the error function,

$$E(t) \leq \phi(t).$$

236 By construction $\phi(t)$ is a continuous function whose extremal values are attained
237 at $\phi(t_i)$ ($i = 0, \dots, m$). Next define

$$G_i := \begin{cases} 0 & \text{if } i = 0, \\ \overline{M}h^{p+1} + G_{i-1}e^{\overline{\mu}\overline{h}} & \text{if } i \geq 1. \end{cases} \quad (18)$$

238 From (17), we see that $\phi(t_i) \leq G_i$ for each i . We claim that

$$G_i = \begin{cases} \overline{M}h^{p+1} & \text{if } \overline{\mu} = 0, \\ \overline{M}h^{p+1} \left(\frac{e^{i\overline{\mu}\overline{h}} - 1}{e^{\overline{\mu}\overline{h}} - 1} \right) & \text{if } \overline{\mu} \neq 0. \end{cases} \quad (19)$$

239 The case $\overline{\mu} = 0$ follows from the formula for $\overline{\mu} \neq 0$ by applying L'Hôpital's rule.
240 Observe that the sequence G_i is non-decreasing with i (even if $\overline{\mu} < 0$). We prove
241 (19) by induction on i . The case $i = 0$ is clear, and for $i \geq 1$:

$$\begin{aligned} G_i &= \overline{M}h^{p+1} + G_{i-1}e^{\overline{\mu}\overline{h}} && \text{(from (18))} \\ &= \overline{M}h^{p+1} + \overline{M}h^{p+1} \left(\frac{e^{(i-1)\overline{\mu}\overline{h}} - 1}{e^{\overline{\mu}\overline{h}} - 1} \right) e^{\overline{\mu}\overline{h}} && \text{(inductive hypothesis for } G_{i-1}) \\ &\vdots \\ &= \overline{M}h^{p+1} \left(\frac{e^{i\overline{\mu}\overline{h}} - 1}{e^{\overline{\mu}\overline{h}} - 1} \right) && \text{(as claimed).} \end{aligned}$$

242 Finally, for any $\delta > 0$, we find an upper bound on $\overline{h}$ in order to ensure that
243 the Taylor curve $T_{q_0, \overline{h}}^p$ stays inside the δ -tube of $\mathbf{x}$. There are 3 cases, depending
244 on $\overline{\mu}$:

245 **CASE 1** ($\overline{\mu} > 0$): The sequence G_i is increasing for $i = 0, \dots, m$, and so it
246 suffices to choose $\overline{h}$ so that $G_m \leq \delta$:

$$\begin{aligned} G_m &= \overline{M}h^{p+1} \left(\frac{e^{m\overline{\mu}\overline{h}} - 1}{e^{\overline{\mu}\overline{h}} - 1} \right) \\ &= \overline{M}h^{p+1} \left(\frac{e^{\overline{\mu}h} - 1}{e^{\overline{\mu}\overline{h}} - 1} \right) && \text{(since } h = \overline{h}m) \\ &\leq \overline{M}h^{p+1} \left(\frac{e^{\overline{\mu}h} - 1}{\overline{\mu}h} \right) && \text{(since } e^x - 1 \geq x \text{ for } x \geq 0) \\ &= \overline{M}h^p \left(\frac{e^{\overline{\mu}h} - 1}{\overline{\mu}} \right). \end{aligned}$$

247 To ensure $G_m \leq \delta$, it suffices to require

$$\begin{aligned} \overline{M}h^p \left(\frac{e^{\overline{\mu}h} - 1}{\overline{\mu}} \right) &\leq \delta \\ \overline{h}^p &\leq \frac{\delta}{\overline{M}} \frac{\overline{\mu}}{(e^{\overline{\mu}\overline{h}} - 1)} \\ \overline{h} &\leq \left(\frac{\delta}{\overline{M}} \frac{\overline{\mu}}{(e^{\overline{\mu}\overline{h}} - 1)} \right)^{1/p}. \end{aligned}$$

248 **CASE 0** ($\bar{\mu} = 0$): This case can be derived from CASE 1 by taking the limit
 249 as $\bar{\mu} \rightarrow 0$, and applying L'Hôpital's rule.

250 **CASE -1** ($\bar{\mu} < 0$): In this case, the above derivation must use this bound
 251 $e^x - 1 < x + \frac{1}{2}x^2$ when $x < 0$. We need some preliminary facts:

- (a) Suppose $\delta > 0$ and $B < B^+ < 0$. Then $\delta > \frac{A}{B^+}$ implies $\delta > \frac{A}{B}$. If $A \geq 0$, the conclusion clearly holds. If $A < 0$ then

$$\delta > \frac{A}{B^+} = \left| \frac{A}{B^+} \right| > \left| \frac{A}{B} \right| = \frac{A}{B}$$

252 since $|B^+| > |B|$.

- 253 (b) Suppose $\bar{\mu} < 0$ and $\bar{h} < 2/|\bar{\mu}|$, then $\bar{\mu} + \frac{1}{2}\bar{\mu}^2\bar{h} < 0$. $\bar{\mu} + \frac{1}{2}\bar{\mu}^2\bar{h} = \bar{\mu}(1 + \frac{1}{2}\bar{\mu}\bar{h}) < \bar{\mu}$
 254 since $0 > \frac{1}{2}\bar{\mu}\bar{h} > -1$.

- (c) It follows from (a) and (b) that to prove that $\delta \geq G_m$, it is sufficient to prove

$$\delta \geq \overline{Mh}^{p+1} \left(\frac{e^{\bar{\mu}h} - 1}{\bar{\mu}\bar{h} + \frac{1}{2}(\bar{\mu}\bar{h})^2} \right) = \overline{Mh}^p \left(\frac{e^{\bar{\mu}h} - 1}{\bar{\mu} + \frac{1}{2}\bar{\mu}^2\bar{h}} \right)$$

255 Rearranging the above inequality, we obtain

$$\bar{h}^p \leq \frac{\delta(\bar{\mu} + \frac{1}{2}\bar{\mu}^2\bar{h})}{\overline{M}(e^{\bar{\mu}h} - 1)} = \frac{\delta\bar{\mu}}{\overline{M}(e^{\bar{\mu}h} - 1)} + \frac{\delta\bar{\mu}^2}{2\overline{M}(e^{\bar{\mu}h} - 1)}\bar{h}.$$

256 Define

$$A := \frac{\delta\bar{\mu}}{\overline{M}(e^{\bar{\mu}h} - 1)}, \quad B := -\frac{\delta\bar{\mu}^2}{2\overline{M}(e^{\bar{\mu}h} - 1)}.$$

257 Since $\bar{\mu} < 0$ and $e^{\bar{\mu}h} - 1 < 0$, both A and B are positive. Hence it is enough
 258 to prove

$$\bar{h}^p + B\bar{h} \leq A.$$

259 Now the function $\bar{h}^p + B\bar{h}$ is increasing for $\bar{h} \geq 0$ since B is positive. There-
 260 fore, if we choose

$$\bar{h} \leq \min \left\{ \left(\frac{A}{2} \right)^{1/p}, \frac{A}{2B} \right\},$$

261 then

$$\bar{h}^p \leq \frac{A}{2}, \quad B\bar{h} \leq \frac{A}{2},$$

262 and thus

$$\bar{h}^p + B\bar{h} \leq A.$$

263 This proves $\delta \geq G_m$.

264 Finally, note that

$$\frac{A}{2B} = \frac{\delta\bar{\mu}/(\overline{M}(e^{\bar{\mu}h} - 1))}{-\delta\bar{\mu}^2/(\overline{M}(e^{\bar{\mu}h} - 1))} = -\frac{1}{\bar{\mu}},$$

265 so our choice also implies $\bar{h} \leq -1/\bar{\mu}$, hence $\bar{\mu} + \frac{1}{2}\bar{\mu}^2\bar{h} < 0$, as required.

266 *Proof of Theorem 1.*

267 Let

$$\mathbf{x}_0 = \text{IVP}(\mathbf{q}_0, h), \quad \mathbf{x} \in \text{IVP}(E_0, h).$$

268 We want to show that

$$\|\mathbf{x}(t) - T_{\mathbf{q}_0, \bar{h}}^p(t)\|_2 \leq \|\mathbf{x}(t) - \mathbf{x}_0(t)\|_2 + \|\mathbf{x}_0(t) - T_{\mathbf{q}_0, \bar{h}}^p(t)\|_2 \leq r_0 e^{\bar{\mu}t} + \delta \quad (20)$$

269 for all $t \in [0, h]$.

270 Hence all hypotheses of Lemma 2 are satisfied, and therefore

$$\|T_{\mathbf{q}_0, \bar{h}}^p(t) - \mathbf{x}_0(t)\| \leq \delta \quad \text{for all } t \in [0, h]. \quad (21)$$

271 Because $\bar{\mu}$ is a lognorm-bound on F_1 , Lemma 1 gives

$$\|\mathbf{x}(t) - \mathbf{x}_0(t)\|_2 \leq e^{\bar{\mu}t} \|\mathbf{p} - \mathbf{q}_0\|_2 \leq r_0 e^{\bar{\mu}t} \quad \text{for all } t \in [0, h]. \quad (22)$$

272 Combining (21) and (22), we obtain (20).

273 (a) Taking $t = h$ in (20), we get

$$\|\mathbf{x}(h) - T_{\mathbf{q}_0, \bar{h}}^p(h)\|_2 \leq r_0 e^{\bar{\mu}h} + \delta.$$

274 Hence

$$\mathbf{x}(h) \in \text{Ball}_{T_{\mathbf{q}_0, \bar{h}}^p(h)}(r_0 e^{\bar{\mu}h} + \delta).$$

275 This ball is an end-enclosure for $\text{IVP}(E_0, h)$.

276 (b) Let

$$r := \max_{h' \in [0, h]} (r_0 e^{\bar{\mu}h'} + \delta).$$

277 Then by (20) for every $t \in [0, h]$,

$$\|\mathbf{x}(t) - T_{\mathbf{q}_0, \bar{h}}^p(t)\|_2 \leq r_0 e^{\bar{\mu}t} + \delta \leq r.$$

278 Therefore,

$$\mathbf{x}(t) \in \text{Ball}_{T_{\mathbf{q}_0, \bar{h}}^p(t)}(r) \subseteq T_{\mathbf{q}_0, \bar{h}}^p(t) \pm [-r, r]^n.$$

279 Taking the union over $t \in [0, h]$, we obtain

$$\mathbf{x}([0, h]) \subseteq T_{\mathbf{q}_0, \bar{h}}^p([0, h]) \pm [-r, r]^n.$$

280 This proves that

$$T_{\mathbf{q}_0, \bar{h}}^p([0, h]) \pm [-r, r]^n$$

281 is a full-enclosure for $\text{IVP}(E_0, h)$.

□

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